Geophysical Research Letters

RESEARCH LETTER
10.1029/2018GL081109

High-Resolution Global Contiguous SIF of OCO-2

L. Yu[1,2,3], J. Wen[1], C. Y. Chang[1], C. Frankenberg[4,5], and Y. Sun[1]

[1]School of Integrative Plant Science, Soil and Crop Sciences Section, Cornell University, Ithaca, NY, USA, [2]Computer Network Information Center, Chinese Academy of Sciences, Beijing, China, [3]School of Computer and Control Engineering, University of Chinese Academy of Sciences, Beijing, China, [4]Division of Geological and Planetary Sciences, California Institute of Technology, Pasadena, CA, USA, [5]Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, USA

Key Points:

- A spatially contiguous global OCO-2 SIF data set at 0.05° and 16-day resolutions ($\overline{\mathrm{SIF}}_{oco2\_005}$) was developed using machine learning and physiological constraints
- $\overline{\mathrm{SIF}}_{oco2\_005}$ successfully preserves the spatiotemporal variability of the original OCO-2 SIF retrievals, captures water stress, and reduces noise
- $\overline{\mathrm{SIF}}_{oco2\_005}$ is highly consistent with independent airborne measurements, demonstrating the effectiveness and validity of the prediction framework

Supporting Information:

- Supporting Information S1

Correspondence to:
Y. Sun,
ys776@cornell.edu

Citation:
Yu, L., Wen, J., Chang, C. Y., Frankenberg, C., & Sun, Y. (2019). High-Resolution Global Contiguous SIF of OCO-2. *Geophysical Research Letters*, *46*, 1449–1458. https://doi.org/10.1029/2018GL081109

Received 26 OCT 2018
Accepted 4 DEC 2018
Accepted article online 11 DEC 2018
Published online 11 FEB 2019

Abstract The Orbiting Carbon Observatory-2 (OCO-2) collects solar-induced chlorophyll fluorescence (SIF) at high spatial resolution along orbits ($\overline{\mathrm{SIF}}_{oco2\_orbit}$), but its discontinuous spatial coverage precludes its full potential for understanding the mechanistic SIF-photosynthesis relationship. This study developed a spatially contiguous global OCO-2 SIF product at 0.05° and 16-day resolutions ($\overline{\mathrm{SIF}}_{oco2\_005}$) using machine learning constrained by physiological understandings. This was achieved by stratifying biomes and times for training and predictions, which accounts for varying plant physiological properties in space and time. $\overline{\mathrm{SIF}}_{oco2\_005}$ accurately preserved the spatiotemporal variations of $\overline{\mathrm{SIF}}_{oco2\_orbit}$ across the globe. Validation of $\overline{\mathrm{SIF}}_{oco2\_005}$ with Chlorophyll Fluorescence Imaging Spectrometer airborne measurements revealed striking consistency ($R^2 = 0.72$; regression slope = 0.96). Further, without time and biome stratification, (1) $\overline{\mathrm{SIF}}_{oco2\_005}$ of croplands, deciduous temperate, and needleleaf forests would be underestimated during the peak season, (2) $\overline{\mathrm{SIF}}_{oco2\_005}$ of needleleaf forests would be overestimated during autumn, and (3) the capability of $\overline{\mathrm{SIF}}_{oco2\_005}$ to detect drought would be diminished.

Plain Language Summary Newly available observations of solar-induced chlorophyll fluorescence (SIF) from satellite sensors represent a major step toward quantifying photosynthesis globally in real time. However, existing satellite SIF records are restricted to low spatial resolutions, sparse data acquisition, or both. These limitations impede the full capability of SIF for improving our understanding of dynamics of photosynthesis and its response to environmental changes (particularly in heterogeneous landscapes) to better support carbon source/sink attribution and verification. This study developed a novel high-resolution time series of spatially contiguous SIF for the globe, leveraging NASA's Orbiting Carbon Observatory-2 measurements. We combined machine learning algorithms with known physiological constraints for this effort. Comparison with independent airborne SIF measurements revealed strong consistency, confirming the high quality of this new SIF data set. The high-resolution and global contiguous coverage of this data set will greatly enhance the synergy between satellite SIF and photosynthesis measured on the ground at consistent spatial scales. Potential applications with this data set include advancing dynamic drought monitoring and mitigation, informing agricultural planning and yield estimation in a more spatially explicit way, and providing a benchmark for upcoming satellite missions with SIF capabilities at higher spatial resolutions.

1. Introduction

Solar-induced chlorophyll fluorescence (SIF) is an optical signal emitted when sunlight is absorbed by chlorophylls within the photosynthetic machinery and therefore provides a functional proxy for photosynthesis (Porcar-Castell et al., 2014). Due to its unique mechanistic relationship with photosynthesis, combined with growing volumes of satellite-based measurements, SIF holds great promise for quantifying global photosynthesis (or gross primary production (GPP; Frankenberg et al., 2011)). Recent advances in satellite SIF retrieval has motivated a number of applications of SIF in ecological, hydrological, and agricultural areas at various spatiotemporal scales (Guanter et al., 2014; Joiner et al., 2014; Liu et al., 2017; Sun et al., 2015, 2017). However, given that SIF itself is a very weak signal (typically <5% of absorbed photons and <2% of the background reflected sunlight, Frankenberg & Berry, 2018) accompanied by comparatively large measurement noise, substantial spatial and temporal averaging are needed to effectively reduce noise, leading to coarse spatial and temporal resolutions of existing satellite SIF products. For example, if aggregated to global uniform grids, the best spatial resolution resolved by Global Ozone Monitoring Experiment-2 (GOME-2) onboard MetOp-A/B is 0.5°, typically at monthly or biweekly scale. Further improvement requires balancing the trade-off between spatial and temporal resolutions to maintain an appropriate noise level. The coarse spatial resolution of existing SIF impedes efforts to obtain a mechanistic understanding of SIF-GPP relationships through robust ground validation, especially in heterogeneous ecosystems. Since October 2017, SIF retrievals became available from the TROPOspheric Monitoring Instrument (TROPOMI; Köhler et al., 2018), which records up to 7×3.5 km^2 pixels with a daily revisit cycle. This product could significantly alleviate the roadblocks of low temporal and spatial resolutions but cannot yet provide long time series of SIF to couple with historical GPP.

Efforts have been made to spatially downscale existing satellite SIF to higher resolutions. For example, Duveiller and Cescatti (2016) developed a framework that empirically links GOME-2 SIF with reflectance-based observations from the MODerate-resolution Imaging Spectroradiometer (MODIS) based on the light use efficiency formulation. This empirical relationship was developed at 0.5° then used to make 0.05° SIF predictions using MODIS data sets at 0.05° as predictors. Gentine and Alemohammad (2018) recently utilized machine learning techniques and MODIS reflectance to reconstruct GOME-2 SIF at 0.5°, which could potentially be applied for downscaling SIF at 0.05°. These efforts confirmed the possibility of creating high-resolution SIF via synergistic use of existing SIF and high-resolution reflectance measurements. However, resolving highly heterogeneous landscapes is still challenging, because the original GOME-2 SIF used to establish the predictive relationship is still at coarse resolution with mixed SIF signals contributed from different land cover types.

The SIF retrievals from NASA's Orbiting Carbon Observatory-2 (OCO-2) (Sun et al., 2018) allows for more spatially explicit studies of SIF, especially in heterogeneous landscapes, due to its high spatial resolution along orbits. For example, a nadir footprint is 1.3 ×2.25 km^2, a size comparable to the footprint of a typical Eddy Covariance (EC) tower. This feature alleviates the issue of scale mismatch between EC-based GPP and satellite SIF from missions such as GOME-2 and enables integration of EC-based GPP with OCO-2 SIF to better understand SIF-GPP relationships (Li, Xiao, & He, 2018; Li, Xiao, He, Altaf Arain, et al., 2018; Smith et al., 2018; Sun et al., 2017; Verma et al., 2017; Wood et al., 2017). However, currently, these efforts are restricted to EC towers that are directly underneath or proximal to OCO-2 overpasses, because OCO-2's high-resolution measurements are made only along narrow orbits, leaving vast spatial gaps between orbits (Figure S1a in the supporting information). As a consequence, the global gridded OCO-2 products are usually generated at 1° grids and monthly scale to ensure spatially contiguous visualization (Sun et al., 2018). OCO-2's data acquisition characteristics thus limit its full potential for elucidating SIF-GPP relationships, constraining global carbon source/sink attribution, and detecting small-scale ecophysiological changes.

Fortunately, OCO-2's acquisition characteristics also create unique opportunities for gap filling of OCO-2 SIF via synergistic exploitation with spatially contiguous, higher spatial and temporal resolution MODIS products. Previously, MODIS products have been used to spatially extrapolate carbon and water fluxes from EC networks to global scales at spatially uniform resolutions (Jung et al., 2009; Tramontana et al., 2016) despite the uneven biome representativeness of EC towers (Schimel et al., 2015). OCO-2 has dense data acquisition and comparatively improved representativeness of diverse biomes and should therefore enable a robust reconstruction of spatially contiguous SIF at high resolutions. This possibility has been confirmed by some initial experiments using localized areas or short time periods (Katzfuss et al., 2018; Li, Xiao, He, Altaf Arain, et al., 2018) and for the globe (Zhang et al., 2018).

This paper aims to (1) develop a novel product of spatially contiguous global 0.05° gridded SIF at 16-day resolution ($\overline{\text{SIF}}_{\text{oco2\_005}}$) for the currently available mission period of OCO-2, that is, from September 2014 to August 2018 and (2) thoroughly validate $\overline{\text{SIF}}_{\text{oco2\_005}}$ with independent airborne SIF measurements from the Chlorophyll Fluorescence Imaging Spectrometer (CFIS). Note that, to the best of our knowledge, this is the first effort so far to validate the value-added high-resolution SIF products with independent SIF measurements, among other studies (Duveiller & Cescatti 2016; Zhang et al., 2018; Gentine & Alemohammad 2018).

2. Materials and Methods

2.1. Prediction Framework

The overall strategy of the prediction framework is to (1) use the artificial neural network (ANN) algorithm (Bishop, 1995) to train the native OCO-2 SIF along orbits against MODIS bidirectional reflectance distribution function (BRDF)-corrected seven-band surface reflectance that are co-located with OCO-2 footprints during

10.1029/2018GL081109

its overpass and (2) apply the trained ANN models to predict SIF in OCO-2's orbital gaps using spatially contiguous MODIS data sets as predictors (Figure S2). This framework is unique and distinct from Zhang et al. (2018) in that the training and prediction were stratified by both biomes and time steps to accurately characterize the dynamic relationship between SIF and reflectance (Damm et al., 2015). SIF contains information about physiological and structural changes of vegetation, while reflectance mainly characterizes structural changes (Porcar-Castell et al., 2014). The biome-specific treatment is designed to capture variations in plant physiology among biomes. The time-specific treatment is enforced to reflect temporal variations in the SIF-reflectance relationship that may occur, for example, when water stress suppresses SIF but does not yet impact reflectance (Daumard et al., 2010). The time-specific treatment enforces SIF as a constraint to optimize the predictive model at each time step.

Based on these rationales, we derived distinct ANN models specific to each biome type at each time step, using feedforward multilayer perceptron architecture (Rumelhart et al., 1986). The number of hidden layers (*nl*) and the number of neurons in each layer (*nn*) were automatically optimized (instead of preset, as in Gentine & Alemohammad, 2018). Optimization was performed using a number of combinations of *nl* (1–3) and *nn* (3–15, odd numbers only to reduce computational cost), and minimized the root-mean-square error (RMSE) using five-fold cross validation, a standard statistical procedure for assessing predictive model performance (Geisser, 2017). The automated determination of *nl* and *nn* attempts to identify the best performing model while minimizing the likelihood of overfitting (Cawley & Talbot, 2010). The optimal *nl* and *nn* combination for each model usually fell within the prescribed boundaries ($nl \leq 3$ and $nn \leq 15$). To ensure even spatial distribution of training and validation samples within each biome (see section 2.2), we divided each biome into sub-biomes by geographical distribution, then extracted data samples within each sub-biome. Specifically, evergreen broadleaf forest was separated into Amazon, Congo, and Southeast Asia; other natural biomes were separated between hemispheres to account for different climate zones; crops were separated among continents to account for distinct crop varieties and management practices (Table S1). For each sub-biome at each time step, we randomly selected 2,000 samples for training and cross validation; this threshold optimized the trade-off between ANN model robustness and computational cost. If the total number of data samples of each time- and biome-specific model were $<$2,000, all were included.

For the final output, 0.05° was chosen as a target resolution, because the random noise of SIF retrievals can be effectively reduced at this resolution via spatial aggregation (detailed in section 2.2), thus facilitating model training and prediction accuracy. The 16-day resolution was chosen because (1) it corresponds to an OCO-2 repeat cycle and (2) MODIS reflectance, although provided daily, are interpolated from a 16-day composite (Schaaf et al., 2002).

2.2. Input Data Sets and Quality Control

The daily mean SIF ($\overline{\text{SIF}}$), converted from instantaneous measurements during OCO-2 overpass, was used to train ANN models. This conversion not only alleviates the impact of varying incoming solar radiation on instantaneous SIF but also generates SIF at a timescale consistent with global GPP (often a temporal integral or mean quantity). SIF normalized by the cosine of the solar zenith angle could also alleviate the effect of varying solar radiation but cannot be directly compared with GPP products unless multiplied by Photosynthetically Active Radiation (PAR) (as in Gentine & Alemohammad, 2018). Among the three viewing geometries of OCO-2 (nadir, glint, and target), we only used nadir mode because it follows a similar ground track during each repeat cycle, allowing for a reliable change detection over time. OCO-2 SIF provides retrievals at 757 and 771 nm. To reduce random noise, this paper used the average of both wavelengths scaled to 757 nm using a weighting factor: $(\text{SIF}_{757} + 1.5\times\text{SIF}_{771})/2$, instead of using SIF_{757} only as in Zhang et al. (2018). Detailed descriptions of the OCO-2 SIF retrieval algorithms, bias correction procedures, daily mean conversion, and wavelength scaling can be found elsewhere (Sun et al., 2018). To ensure ANN model accuracy, OCO-2 SIF pixels were considered valid and extracted only if they satisfied data quality criteria defined by Sun et al. (2018), that is, excluding unsuccessful retrievals, large solar zenith angle, and clouds. The five nearest valid OCO-2 footprints of the same land cover type, approximating the target resolution 0.05°, were aggregated to reduce random noise in individual footprints. According to Frankenberg et al. (2014), random retrieval errors are reduced by $\sqrt{n}$, where *n* is the number of footprints included for aggregation. Thus, using the five nearest OCO-2 footprints could effectively reduce random noise by ~2.3. Lastly, the aggregated pixels were used for ANN model training (described in section 2.1).

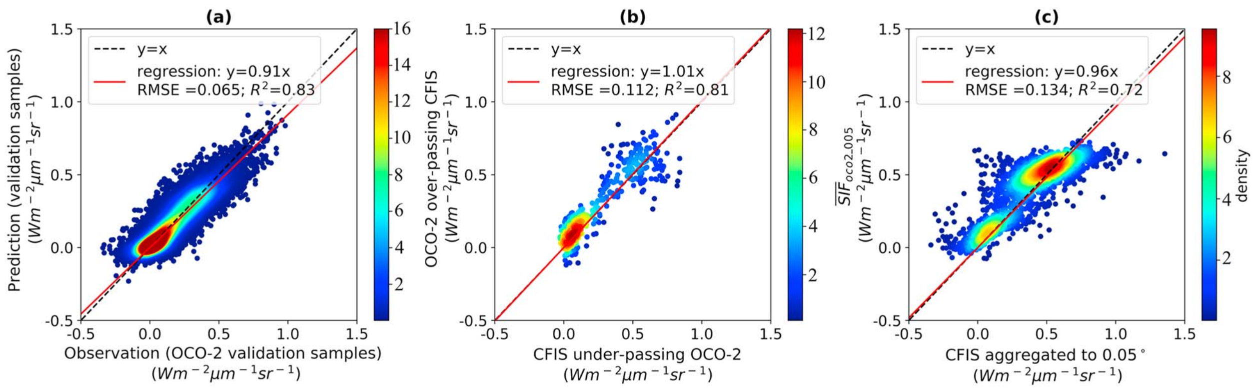

Figure 1. Validation of the prediction framework. (a) Predicted (*y* axis) versus observed (*x* axis) OCO-2 SIF along orbits using five-fold cross validation from September 2014 to August 2018. For figure clarity, only 50,000 randomly selected samples are displayed. (b) OCO-2 SIF along orbits versus coordinated CFIS flights under-passing OCO-2 orbits. The OCO-2 SIF along orbits in (a) and (b) refers to $\overline{SIF}_{oco2\_orbit}$ aggregated from five nearest valid pixels of the same land cover type. The CFIS footprints under-passing OCO-2 orbits in (b) were spatially aggregated to match footprints of the aggregated $\overline{SIF}_{oco2\_orbit}$. (c) $\overline{SIF}_{oco2\_005}$ versus CFIS SIF (aggregated to 0.05°) in areas that did not under-pass OCO-2 orbits. OCO-2 = Orbiting Carbon Observatory-2; SIF = solar-induced chlorophyll fluorescence; CFIS = Chlorophyll Fluorescence Imaging Spectrometer; RMSE = root-mean-square error.

The MODIS BRDF-corrected seven-band surface reflectance comes from MCD43A4 and MCD43C4 V006 products. Both are generated daily with the same retrieval algorithms but delivered at different spatial resolutions (Schaaf & Wang, 2015). MCD43A4 at 500-m resolution was used for model training for each biome along OCO-2 orbits, which can account for subgrid heterogeneity within a 0.05° grid. The daily MCD43C4 data set at 0.05° resolution was aggregated to 16 days for prediction at the global scale. This is different from Zhang et al. (2018), who used MCD43C4 at 0.05° for both model training and prediction. MODIS pixels labeled with high-quality assurance (QA = 0) flags for all seven bands were considered valid pixels and extracted to construct training and validation samples. Valid MODIS pixels co-located with aggregated OCO-2 pixels of the same land cover type were also aggregated to match their spatial coverage.

We used the MCD12Q1 (500 m) and MCD12C1 (0.05°) land cover products (V005) with the IGBP (International Geosphere-Biosphere Programme) classification scheme (Friedl et al., 2002), for training and prediction, respectively. For simplicity, we combined certain IGBP types into a smaller number of classes following Frankenberg et al. (2011; Figure S3).

2.3. Validation Approaches

We evaluated the prediction framework using three approaches. First, we used five-fold cross validation to compare predicted $\overline{SIF}$ along OCO-2 orbits against observations (aggregated from five nearest pixels) that were not used in training, that is, 20% of samples selected for cross validation for each sub-biome and time step during the entire period. Second, we compared predicted $\overline{SIF}$ with independent airborne measurements from CFIS, which was specifically designed by the OCO-2 team to validate OCO-2 SIF retrieval (Frankenberg et al., 2018). This study used all CFIS campaigns conducted in 2016 to validate $\overline{SIF}_{oco2\_005}$ where it was co-located with CFIS flights. To achieve this, we spatially aggregated CFIS SIF to match the 0.05° resolution of $\overline{SIF}_{oco2\_005}$. CFIS retrieves SIF at 755 nm, which is roughly 12% higher in magnitude than OCO-2's 757 nm (van der Tol et al., 2014). We thus applied a 1/1.12 scaling factor to CFIS SIF prior to comparison with $\overline{SIF}_{oco2\_005}$. Third, we reaggregated $\overline{SIF}_{oco2\_005}$ to 1° and examined its residual from the original OCO-2 SIF gridded at 1°. This cross-check confirms the preservation of spatiotemporal variability after spatial extrapolation if the residual is randomly distributed with a zero mean. We used a suite of statistical metrics to evaluate the prediction, including RMSE, coefficient of determination (R^2), and slope determined from ordinary least squares regression.

3. Results and Discussion

3.1. Validation of Prediction Framework

Figure 1a demonstrates the effectiveness of our prediction framework using the first validation approach described in section 2.3. When pooling together all biomes and time steps, the prediction statistics are:

R^2 = 0.83, RMSE = 0.065 W $m^{-2} \cdot \mu m^{-1} \cdot sr^{-1}$, and slope = 0.91 (Figure 1a). However, prediction performance varied with biomes. Using August 2015 as an example (Figure S4), $R^2 > 0.75$ and RMSE $\leq$ 0.075 W $m^{-2} \cdot \mu m^{-1} \cdot sr^{-1}$ were observed for croplands (CRO), deciduous broadleaf forests (DBF), and savannas (SAV). The regression slopes were close to one for these biomes as well as needleleaf forests (NF) and evergreen broadleaf forests (EBF). However, both shrubland (SHR) and grassland (GRA) under-predicted high values and overpredicted low values, although the mean value was well predicted (Figure S4). The weakened fitting performance may result from the lower magnitude of SIF and consequently higher impact of noise in GRA and SHR relative to more productive biomes. Another possibility is that highly variable precipitation for GRA and SHR (Beringer et al., 2016) could trigger rapid changes in plant productivity that might be masked at a 16-day temporal resolution. Evaluation metrics were similar between validation (Figure S4) and training data sets (Figure S5) for all biomes, confirming minimal overfitting of our optimized ANN models. We further demonstrated that averaging the two OCO-2 SIF bands improved prediction performance by reducing random noise (Table S2).

We further compared $\overline{SIF}_{oco2\_005}$ with independent SIF measurements from CFIS campaigns that surveyed both natural vegetation and managed systems. We first performed a baseline comparison between native OCO-2 SIF along orbits and coordinated CFIS flights under-passing OCO-2 (R^2 = 0.81, slope = 1.01; Figure 1b), to confirm that there was no systematic discrepancy in the native SIF retrievals. We then compared predicted $\overline{SIF}_{oco2\_005}$ that was co-located with all CFIS flights (CFIS aggregated to 0.05° grids) that were not under-passing OCO-2 orbits (Figure 1c). The R^2 value of 0.72 demonstrates that our prediction framework can effectively capture SIF variability in regions without OCO-2 overpasses. The regression slope of 0.96 indicates small prediction bias in $\overline{SIF}_{oco2\_005}$.

A global evaluation revealed that $\overline{SIF}_{oco2\_005}$ accurately captured the spatial variations in SIF depicted by the original OCO-2 retrieval (Figure S6). Using August 2015 as an example, we cross-checked $\overline{SIF}_{oco2\_005}$ reaggregated to 1° against $\overline{SIF}_{oco2\_1x1}$ ($\overline{SIF}_{oco2\_orbit}$ aggregated to 1°) and found minimal systematic differences (Figure S6a), with prediction residuals ($\overline{SIF}_{oco2\_005} - \overline{SIF}_{oco2\_1x1}$) randomly distributed in space (except for stripe-like patterns) and a near-zero global mean (Figure S6b). Note that the striped pattern in Figure S6a is not a bias present in $\overline{SIF}_{oco2\_005}$ but instead arose from propagation of stripes present in $\overline{SIF}_{oco2\_1x1}$ due to spatial gaps in the original $\overline{SIF}_{oco2\_orbit}$ (Figure S7). The pattern of minimal prediction residuals held throughout the study period (Figure S6c). Overall, our results indicate that both spatial and temporal variabilities of the original SIF retrievals are preserved in $\overline{SIF}_{oco2\_005}$.

3.2. Spatial and Temporal Patterns of $\overline{SIF}_{oco2\_005}$

Despite the limited number of orbits in each 16-day cycle (Figure S1a), $\overline{SIF}_{oco2\_005}$ was able to capture global spatial variations exhibited in the original OCO-2 $\overline{SIF}$ (Figure 2a versus Figures S1b and S1c). For example, in August 2015, the highest SIF was observed in agricultural sectors such as the U.S. Midwestern corn belt, which even exceeded the SIF magnitude in the pan-tropics. Moderate SIF was observed in DBF and high-latitude NF, while low SIF was observed in arid or nonforested areas in the Southern Hemisphere during its austral winter. These patterns are consistent with the spatial distribution of GPP reported by previous studies (e.g., Beer et al., 2010). Notably, the high productivity of the U.S. Midwestern corn belt, the rice belt along the lower Mississippi river, and the California central valley are uniquely identified by $\overline{SIF}_{oco2\_005}$ (Figure 2b), whereas MODIS EVI poorly distinguishes these agricultural regions from temperate deciduous forests (Figure 2d). $\overline{SIF}_{oco2\_005}$ also successfully identifies isolated cropping systems, for example, barley in Idaho and cotton in northwestern Texas, which cannot be resolved by the original OCO-2 SIF at 1° (Figure 2b vs. 2c).

The temporal variation in $\overline{SIF}_{oco2\_005}$ closely resembles that of the original SIF retrievals, both along the orbits ($\overline{SIF}_{oco2\_orbit}$) and gridded at 1° ($\overline{SIF}_{oco2\_1x1}$), for most biomes and regions during the OCO-2 mission period (Figure 3). Their striking consistency holds under both normal and stressed conditions. For example, $\overline{SIF}_{oco2\_005}$ closely followed $\overline{SIF}_{oco2\_orbit}$ from the dry to wet season (Figure 3a, July to November) during a severe 2015 El Niño event in the Amazon basin. In particular, the decrease of $\overline{SIF}_{oco2\_orbit}$ during peak drought in November 2015 (Koren et al., 2018) was successfully captured by $\overline{SIF}_{oco2\_005}$ (Figure S8b). Notably, the standard deviation of the mean for $\overline{SIF}_{oco2\_005}$ is considerably reduced relative to $\overline{SIF}_{oco2\_orbit}$, indicating

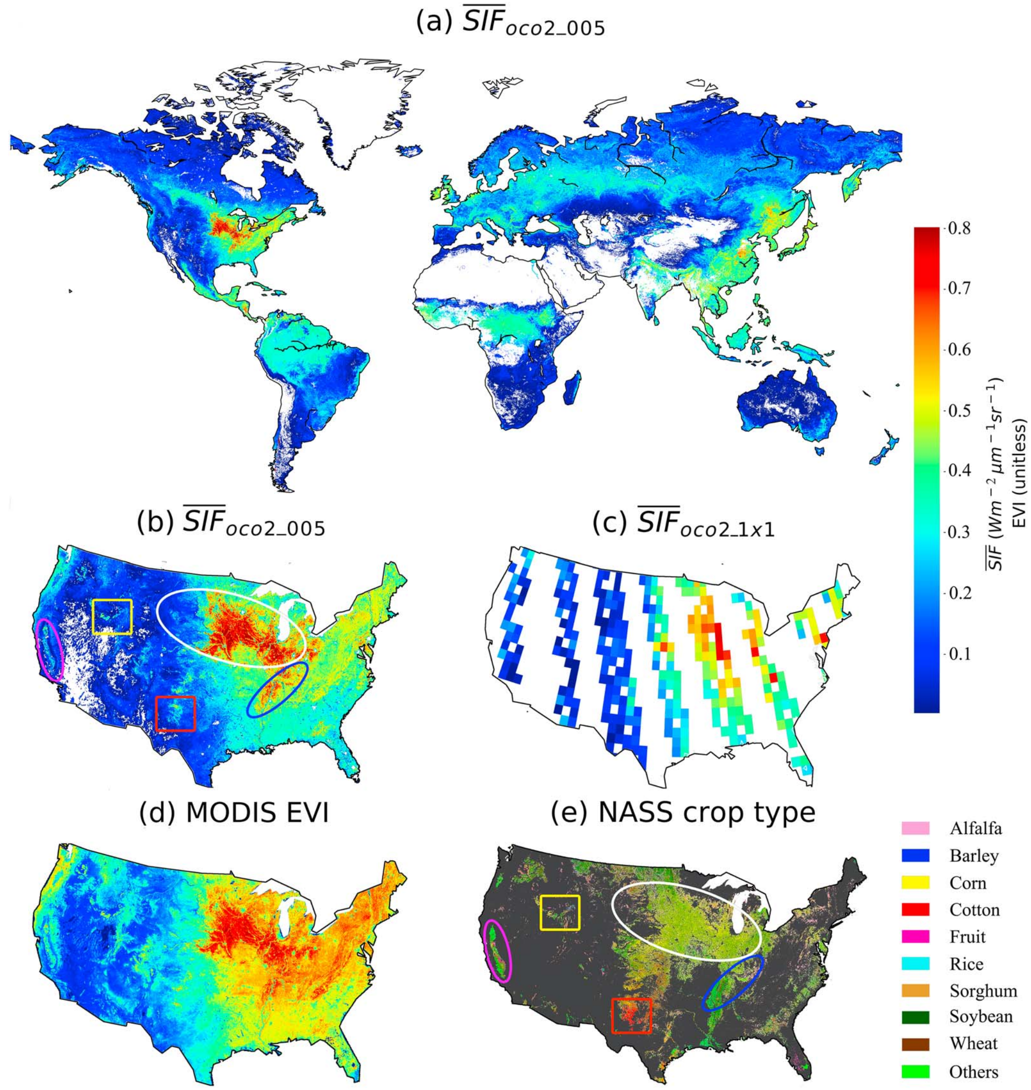

Figure 2. Predicted spatially contiguous $\overline{\mathrm{SIF}}_{\mathrm{oco2\_005}}$ in the first 16 days of August 2015 for (a) the globe and (b) United States; (c) $\overline{\mathrm{SIF}}_{\mathrm{oco2\_orbit}}$ aggregated to 1° grid and (d) MODIS EVI during the same period, where 16-day EVI is calculated from the MODIS daily BRDF-corrected surface reflectance; (e) United States Department of Agriculture National Agricultural Statistics Service (NASS) cropland data layer in 2015. Ovals in (c) and (e) highlight the U.S. Midwestern corn belt (white), the rice belt along the lower Mississippi river (blue), and the California central valley (pink). Boxes highlight production areas for barley in Idaho (yellow) and cotton in northwestern Texas (red). SIF = solar-induced chlorophyll fluorescence; MODIS = MODerate-resolution Imaging Spectroradiometer; EVI = enhanced vegetation index.

decreased noise in the predicted $\overline{\mathrm{SIF}}_{\mathrm{oco2\_005}}$. We observed some discrepancies in temporal variation between $\overline{\mathrm{SIF}}_{\mathrm{oco2\_005}}$ and original retrievals. For example, slight offsets of $\overline{\mathrm{SIF}}_{\mathrm{oco2\_005}}$ from original retrievals in the Congo basin from July to August 2015 (Figure 3b) and in the Amazon basin from November to December 2016 (Figure 3a) were likely due to missing valid MODIS pixels in these regions that led to spatial gaps in $\overline{\mathrm{SIF}}_{\mathrm{oco2\_005}}$ prediction, although there were OCO-2 overpasses (Figures S8a vs. S8b). Prediction gaps existed (Figure 3), either because OCO-2 did not provide SIF measurements (August to September 2017) or had very limited orbital coverage (April 2015, Figure S9). Overall, $\overline{\mathrm{SIF}}_{\mathrm{oco2\_005}}$ exhibited minimal prediction bias and preserved temporal variability of original retrievals.

3.3. Benefits of Time- and Biome-Specific Model

To demonstrate the benefits of invoking the time- and biome-specific strategy in our prediction framework, we performed four experiments: a universal model, a time-specific model without biome stratification, a

10.1029/2018GL081109

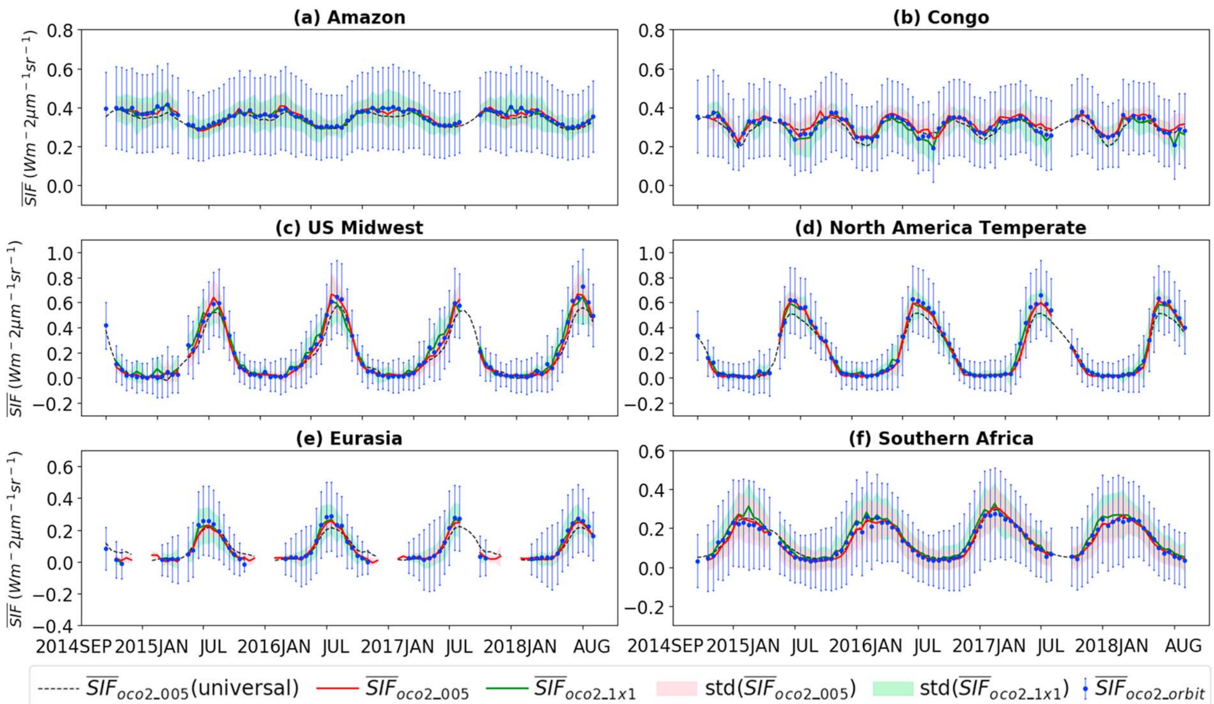

Figure 3. Time series of $\overline{\text{SIF}}_{\text{oco2\_005}}$, $\overline{\text{SIF}}_{\text{oco2\_orbit}}$, and $\overline{\text{SIF}}_{\text{oco2\_1x1}}$ for selected regions across the globe (boundaries identical to Sun et al., 2018). Blue error bars represent standard deviation across original footprints; red and green shading represent standard deviation across 0.05° and 1° grids, respectively. Black dashed line indicates $\overline{\text{SIF}}_{\text{oco2\_005}}$ predicted from the universal model; standard deviation is not shown for figure clarity. SIF = solar-induced chlorophyll fluorescence.

biome-specific model without time stratification, and the time- and biome-specific model developed in this study. Models without time stratification used all data within 2015–2016 for training. Note here that we did not attempt to compare our products directly with Zhang et al. (2018) who used a universal model, because different procedures for pre-processing SIF (i.e., wavelength averaging to reduce noise, 500m land cover data for training to account for sub-grid heterogeneity, details provided above) may confound interpreting the benefits of time-and-biome-specific model. The four experiments we performed here ensures a fair comparison by making SIF pre-processing identical with each other. Using $\overline{\text{SIF}}_{\text{oco2\_005}}$ predictions in August 2015 as an example, we found that the time- and biome-specific framework outperformed the universal model for all biomes except grassland in terms of R^2 and RMSE (Table S3). In particular, regression slopes and RMSE for croplands (using pixels from North America as an example) consistently improved from universal (lowest performance) to time-specific, biome-specific, and time- and biome-specific (best performance) models, though all four experiments had a similar R^2 (~0.76, Figure S10). The universal model tended to predict saturated SIF when the true SIF observation was high (Figure S10a), which likely propagated from MODIS reflectance which often saturates under high productivity. This saturation effect was greatly alleviated when using the time- and biome-specific model (Figure S10d). Indeed, combined time and biome stratification considerably outperformed the universal model when validated with CFIS campaigns (Figures 1c vs. S11).

Global analyses further revealed the consequences of not stratifying time steps and biomes (Figure S12). In the peak growing season (June to July 2015) of the Northern Hemisphere, the universal model consistently yielded lower $\overline{\text{SIF}}_{\text{oco2\_005}}$ values than our time- and biome-specific framework for CRO, DBF, and NF. However, during autumn (September to October), the universal model tended to overpredict SIF in NF. These spatial patterns are consistent with the time series shown in Figures 3c–3e.

The time- and biome-specific strategy also outperformed the universal model when characterizing water stress. For example, during the 2015 drought in the U.S. western coast (Figure S13a), $\overline{\text{SIF}}_{\text{oco2\_005}}$ predicted from the time- and biome-specific model showed a widespread decline (Figure 13c) relative to 2016, a moderate dry year (Figure 13b), consistent with the progressive canopy water loss reported by Asner et al. (2016).

However, little difference between the two years was found using the universal model (Figure S13d), highlighting the advantage of time and biome stratification for detecting water stress.

These benefits are likely due to the physiological constraints that time and biome stratification imposes onto machine learning. A universal model directly carries over variations in reflectance to predicted SIF but does not account for physiological variations (e.g., SIF yield) that occur over time and among biomes (Damm et al., 2015). In contrast, a time- and biome-specific model is able to adjust the SIF-reflectance relationship, and hence implicitly accounts for physiological variations in SIF yield in the ANN model.

3.4. Caveats and Potential New Applications With $\overline{\mathrm{SIF}}_{oco2\_005}$

We envision that the new $\overline{\mathrm{SIF}}_{oco2\_005}$ product can greatly advance SIF application in multiple aspects. First, its high-resolution and contiguous coverage will improve synergistic use of OCO-2 SIF with GPP from EC towers, such as Fluxnet and Long-Term Agroecosystem Research (LTAR) networks, facilitating mechanistic investigation of SIF-GPP relationships. Li, Xiao, and He (2018), Li, Xiao, He, Altaf Arain, et al. (2018), and Sun et al. (2017) conducted such investigations but used a limited number of EC sites, because the concurrent availability and consistent spatial footprints between OCO-2 SIF and EC sites were strongly limited by OCO-2's orbital gaps. $\overline{\mathrm{SIF}}_{oco2\_005}$ can greatly increase inclusion of available EC GPP for such endeavors. Second, $\overline{\mathrm{SIF}}_{oco2\_005}$ can serve as a benchmark for other satellite SIF products from TROPOMI and upcoming missions such as OCO-3, Sentinel-4/5, Tropospheric Emissions Monitoring of POllution (TEMPO; Zoogman et al., 2017), FLuorescence EXplorer (FLEX; Drusch et al., 2017), and Geostationary Carbon Observatory (GeoCARB; Polonsky et al., 2014), which have SIF capabilities at higher spatial and/or temporal resolutions than existing missions such as GOME-2. Third, $\overline{\mathrm{SIF}}_{oco2\_005}$ can help advance dynamic drought monitoring and mitigation as well as inform agricultural planning and yield estimation in a more spatially explicit way (Guanter et al., 2014).

Nevertheless, future work is needed to further improve the reliability of $\overline{\mathrm{SIF}}_{oco2\_005}$. In particular, predictive uncertainty in orbital gaps should be evaluated, despite the initial consistency found between predicted $\overline{\mathrm{SIF}}_{oco2\_005}$ and CFIS. Four major sources of uncertainties in $\overline{\mathrm{SIF}}_{oco2\_005}$ warrant caution in applying and interpreting this product. First, retrieval errors in the original OCO-2 SIF and MODIS measurements may propagate into ANN training and prediction. Though quality flags provided by MODIS products can serve as initial screening criteria, missing values due to cloud or aerosol contamination can create substantial data gaps in prediction, even when original SIF retrievals are available. Second, despite improved biome representativeness of OCO-2 orbits compared to EC tower distribution, uneven training sampling occurs for different biomes at different time steps due to sparse valid observations of original SIF and instrument issues of OCO-2 (e.g., April 2015). This may create differential predictability of ANN models. Data gaps in original OCO-2 SIF could also lead to prediction gaps of $\overline{\mathrm{SIF}}_{oco2\_005}$. This is a caveat of time stratification; a universal approach is not constrained by data gaps in the original SIF as long as predictors are available. Third, predictive uncertainties may arise from classification errors in land cover products, especially in highly heterogeneous landscapes and heavily human-managed systems. Indeed, Xiao et al. (2011) demonstrated that the 30 m resolution National Land Cover Data (Homer et al., 2015) exhibited considerable differences in distribution and areas of biome coverage from MODIS products in the United States, which greatly impacted spatial extrapolation of carbon fluxes from EC towers to the regional scale. Further improvement in the predictive performance of $\overline{\mathrm{SIF}}_{oco2\_005}$ requires resolving subgrid spatial variation in heterogeneous landscapes. In particular, C3/C4 differentiation warrants further investigation with higher resolution C3/C4 resolved land cover products, which was not addressed here because the MODIS data set does not distinguish these two photosynthetic pathways. Finally, uncertainties may also originate from the assumption of consistent plant activity within a 16-day period, which is especially unlikely during phenological transition periods. This uncertainty may be evaluated with TROPOMI during their overlapping period.

4. Conclusions

This study developed a global 16-day spatially contiguous OCO-2 SIF product at 0.05° using a machine learning approach trained to account for physiological differences. We employed ANN to train the native OCO-2 SIF observations with MODIS BRDF-corrected seven-band surface reflectance along OCO-2's orbits. The trained predictive models were then applied to predict SIF in OCO-2's gap regions using MODIS data sets as predictors. This framework was stratified by biomes and time steps. Our results showed that the

10.1029/2018GL081109

developed $\overline{SIF}_{oco2\_005}$ accurately captured spatiotemporal patterns of the original OCO-2 SIF across the globe, suggesting that the spatial and temporal variability of original SIF is preserved. Comparison of $\overline{SIF}_{oco2\_005}$ with independent CFIS airborne measurements showed high consistency, confirming the effectiveness of the developed framework. We further demonstrated that without stratifying time and biomes, the peak SIF magnitudes of CRO, DBF, and NF would be underpredicted, while SIF for NF during autumn would be overpredicted. In addition, time and biome stratification could improve water stress detection. Finally, we offer insights for limitations, potential improvement, and applications of this new product.

Acknowledgments
L. Y. acknowledges 2018 CSC-IBM Future Data Scientist Scholarship Program; C. Y. C. acknowledges USDA-NIFA postdoctoral fellowship 2018-67012-27985. C. F. acknowledges OCO-2 science team 80NSSC18K0895, Carbon Cycle Science NNX17AE14G, and JPL support for CFIS. Y. S. and J. W. acknowledge the Earth Science Division MEaSUREs program. Data sources: MODIS https://e4ftl01.cr.usgs.gov/MOTA/, OCO-2 SIF https://co2.jpl.nasa.gov, CFIS SIF ftp://fluo.gps.caltech.edu/data/CFIS, $\overline{SIF}_{oco2\_005}$ https://cornell.app.box.com/s/cavtg50y80udbdirg022gm5whugmth02.

References

Asner, G. P., Brodrick, P. G., Anderson, C. B., Vaughn, N., Knapp, D. E., & Martin, R. E. (2016). Progressive forest canopy water loss during the 2012–2015 California drought. *Proceedings of the National Academy of Sciences*, *113*(2), E249–E255.

Beer, C., Reichstein, M., Tomelleri, E., Ciais, P., Jung, M., Carvalhais, N., et al. (2010). Terrestrial gross carbon dioxide uptake: Global distribution and covariation with climate. *Science*, *329*(5993), 834–838. https://doi.org/10.1126/science.1184984

Beringer, J., Hutley, L. B., McHugh, I., Arndt, S. K., Campbell, D., Cleugh, H. A., et al. (2016). An introduction to the Australian and New Zealand flux tower network – OzFlux. *Biogeosciences*, *13*(21), 5895–5916. https://doi.org/10.5194/bg-13-5895-2016

Bishop, C. M. (1995). *Neural networks for pattern recognition*. Oxford: Oxford University Press.

Cawley, G. C., & Talbot, N. L. (2010). On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, *58*(4), 264–267. https://doi.org/10.1016/j.biopha.2003.08.027

Damm, A., Guanter, L., Paul-Limoges, E., van der Tol, C., Hueni, A., Buchmann, N., et al. (2015). Far-red sun-induced chlorophyll fluorescence shows ecosystem-specific relationships to gross primary production: An assessment based on observational and modeling approaches. *Remote Sensing of Environment*, *166*, 91–105. https://doi.org/10.1016/j.rse.2015.06.004

Daumard, F., Champagne, S., Fournier, A., Goulas, Y., Ounis, A., Hanocq, J.-F., & Moya, I. (2010). A field platform for continuous measurement of canopy fluorescence. *IEEE Transactions on Geoscience and Remote Sensing*, *48*(9), 3358–3368. https://doi.org/10.1109/TGRS.2010.2046420

Drusch, M., Moreno, J., Del Bello, U., Franco, R., Goulas, Y., Huth, A., et al. (2017). The FLuorescence EXplorer Mission concept—ESA's Earth Explorer 8. *IEEE Transactions on Geoscience and Remote Sensing*, *55*(3), 1273–1284. https://doi.org/10.1109/TGRS.2016.2621820

Duveiller, G., & Cescatti, A. (2016). Spatially downscaling sun-induced chlorophyll fluorescence leads to an improved temporal correlation with gross primary productivity. *Remote Sensing of Environment*, *182*, 72–89. https://doi.org/10.1016/j.rse.2016.04.027

Frankenberg, C., Köhler, P., Magney, T. S., Geier, S., Lawson, P., Schwochert, M., et al. (2018). The Chlorophyll Fluorescence Imaging Spectrometer (CFIS), mapping far red fluorescence from aircraft. *Remote Sensing of Environment*, *217*, 523–536.

Frankenberg, C., & Berry, J. (2018). Solar induced chlorophyll fluorescence: Origins, relation to photosynthesis and retrieval. *Comprehensive Remote Sensing*. https://doi.org/10.1016/B978-0-12-409548-9.10632-3

Frankenberg, C., Fisher, J. B., Worden, J., Badgley, G., Saatchi, S. S., Lee, J.-E., et al. (2011). New global observations of the terrestrial carbon cycle from GOSAT: Patterns of plant fluorescence with gross primary productivity. *Geophysical Research Letters*, *38*, L17706. https://doi.org/10.1029/2011GL048738

Frankenberg, C., O'Dell, C., Berry, J., Guanter, L., Joiner, J., Köhler, P., et al. (2014). Prospects for chlorophyll fluorescence remote sensing from the Orbiting Carbon Observatory-2. *Remote Sensing of Environment*, *147*, 1–12. https://doi.org/10.1016/j.rse.2014.02.007

Friedl, M. A., McIver, D. K., Hodges, J. C. F., Zhang, X. Y., Muchoney, D., Strahler, A. H., et al. (2002). Global land cover mapping from MODIS: Algorithms and early results. *Remote Sensing of Environment*, *83*(1–2), 287–302. https://doi.org/10.1016/S0034-4257(02)00078-0

Geisser, S. (2017). Predictive inference: An introduction. https://doi.org/10.1201/9780203742310

Gentine, P., & Alemohammad, S. H. (2018). Reconstructed solar-induced fluorescence: A machine learning vegetation product based on MODIS surface reflectance to reproduce GOME-2 solar-induced fluorescence. *Geophysical Research Letters*, *45*, 3136–3146. https://doi.org/10.1002/2017GL076294

Guanter, L., Zhang, Y., Jung, M., Joiner, J., Voigt, M., Berry, J. A., et al. (2014). Global and time-resolved monitoring of crop photosynthesis with chlorophyll fluorescence. *Proceedings of the National Academy of Sciences of the United States of America*, *111*(14), E1,327–E1,333. https://doi.org/10.1073/pnas.1320008111

Homer, C. G., Dewitz, J. A., Yang, L., Jin, S., Danielson, P., Xian, G., et al. (2015). Completion of the 2011 National Land Cover Database for the conterminous United States-representing a decade of land cover change information. *Photogrammetric Engineering and Remote Sensing*, *81*(5), 345–354.

Joiner, J., Yoshida, Y., Vasilkov, A. P., Schaefer, K., Jung, M., Guanter, L., et al. (2014). The seasonal cycle of satellite chlorophyll fluorescence observations and its relationship to vegetation phenology and ecosystem atmosphere carbon exchange. *Remote Sensing of Environment*, *152*, 375–391. https://doi.org/10.1016/j.rse.2014.06.022

Jung, M., Reichstein, M., & Bondeau, A. (2009). Towards global empirical upscaling of FLUXNET eddy covariance observations: Validation of a model tree ensemble approach using a biosphere model. *Biogeosciences*, *6*(10), 2001–2013. https://doi.org/10.5194/bg-6-2001-2009

Katzfuss, M., Guinness, J., & Gong, W. (2018). Vecchia approximations of Gaussian-process predictions. Retrieved from http://arxiv.org/abs/1805.03309

Köhler, P., Frankenberg, C., Magney, T. S., Guanter, L., Joiner, J., & Landgraf, J. (2018). Global retrievals of solar-induced chlorophyll fluorescence with TROPOMI: First results and Intersensor comparison to OCO-2. *Geophysical Research Letters*, *45*, 10,456–10,463. https://doi.org/10.1029/2018GL079031

Koren, G., van Schaik, E., Araújo, A. C., Boersma, K. F., Gärtner, A., Killaars, L., et al. (2018). Widespread reduction in sun-induced fluorescence from the Amazon during the 2015/2016 El Niño. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *373*(1760). Retrieved from http://rstb.royalsocietypublishing.org/content/373/1760/20170408.abstract), https://doi.org/10.1098/rstb.2017.0408

Li, X., Xiao, J., & He, B. (2018). Chlorophyll fluorescence observed by OCO-2 is strongly related to gross primary productivity estimated from flux towers in temperate forests. *Remote Sensing of Environment.*, *204*, 659–671. https://doi.org/10.1016/j.rse.2017.09.034

Li, X., Xiao, J., He, B., Altaf Arain, M., Beringer, J., Desai, A. R., et al. (2018). Solar-induced chlorophyll fluorescence is strongly correlated with terrestrial photosynthesis for a wide variety of biomes: First global analysis based on OCO-2 and flux tower observations. *Global Change Biology*, *24*(9), 3990–4008. https://doi.org/10.1111/gcb.14297

Liu, J., Bowman, K. W., Schimel, D. S., Parazoo, N. C., Jiang, Z., Lee, M., et al. (2017). Contrasting carbon cycle responses of the tropical continents to the 2015–2016 El Niño. *Science*, *358*(6360), eaam5690.

10.1029/2018GL081109

Polonsky, I. N., O'Brien, D. M., Kumer, J. B., O'Dell, C. W., & the geoCARB Team (2014). Performance of a geostationary mission, geoCARB, to measure CO_2, CH_4 and CO column-averaged concentrations. *Atmospheric Measurement Techniques*, *7*(4), 959–981. https://doi.org/10.5194/amt-7-959-2014

Porcar-Castell, A., Tyystjärvi, E., Atherton, J., Van Der Tol, C., Flexas, J., Pfündel, E. E., et al. (2014). Linking chlorophyll a fluorescence to photosynthesis for remote sensing applications: Mechanisms and challenges. *Journal of Experimental Botany*, *65*, 4065–4095.

Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, *323*(6088), 533–536. https://doi.org/10.1038/323533a0

Schaaf, C. B., Gao, F., Strahler, A. H., Lucht, W., Li, X., Tsang, T., et al. (2002). First operational BRDF, albedo nadir reflectance products from MODIS. *Remote Sensing of Environment*, *83*(1-2), 135–148. https://doi.org/10.1016/S0034-4257(02)00091-3

Schaaf, C., & Wang, Z. (2015). MCD43A3 MODIS/Terra+Aqua BRDF/Albedo Daily L3 Global - 500m V006 [Data set]. NASA EOSDIS Land Processes DAAC. https://doi.org/10.5067/MODIS/MCD43A3.006

Schimel, D., Pavlick, R., Fisher, J. B., Asner, G. P., Saatchi, S., Townsend, P., et al. (2015). Observing terrestrial ecosystems and the carbon cycle from space. *Global Change Biology*, *21*, 1762–1776.

Smith, W. K., Biederman, J. A., Scott, R. L., Moore, D. J. P., He, M., Kimball, J. S., et al. (2018). Chlorophyll fluorescence better captures seasonal and interannual gross primary productivity dynamics across dryland ecosystems of southwestern North America. *Geophysical Research Letters*, *45*, 748–757. https://doi.org/10.1002/2017GL075922

Sun, Y., Frankenberg, C., Jung, M., Joiner, J., Guanter, L., Köhler, P., & Magney, T. (2018). Overview of solar-induced chlorophyll fluorescence (SIF) from the Orbiting Carbon Observatory-2: Retrieval, cross-mission comparison, and global monitoring for GPP. Remote Sensing of Environment.

Sun, Y., Frankenberg, C., Wood, J. D., Schimel, D. S., Jung, M., Guanter, L., et al. (2017). OCO-2 advances photosynthesis observation from space via solar-induced chlorophyll fluorescence. *Science*, *358*(6360), eaam5747. https://doi.org/10.1126/science.aam5747

Sun, Y., Fu, R., Dickinson, R., Joiner, J., Frankenberg, C., Gu, L., et al. (2015). Drought onset mechanisms revealed by satellite solar-induced chlorophyll fluorescence: Insights from two contrasting extreme events. *Journal of Geophysical Research G: Biogeosciences*, *120*, 2427–2440. https://doi.org/10.1002/2015JG003150

Tramontana, G., Jung, M., Schwalm, C. R., Ichii, K., Camps-Valls, G., Ráduly, B., et al. (2016). Predicting carbon dioxide and energy fluxes across global FLUXNET sites with regression algorithms. *Biogeosciences*, *13*(14), 4291–4313. https://doi.org/10.5194/bg-13-4291-2016

Van Der Tol, C., Berry, J. A., Campbell, P. K. E., & Rascher, U. (2014). Models of fluorescence and photosynthesis for interpreting measurements of solar-induced chlorophyll fluorescence. *Journal of Geophysical Research: Biogeosciences*, *119*, 2312–2327. https://doi.org/10.1002/2014JG002713

Verma, M., Schimel, D., Evans, B., Frankenberg, C., Beringer, J., Drewry, D. T., et al. (2017). Effect of environmental conditions on the relationship between solar-induced fluorescence and gross primary productivity at an OzFlux grassland site. *Journal of Geophysical Research: Biogeosciences*, *122*, 716–733. https://doi.org/10.1002/2016JG003580

Wood, J. D., Griffis, T. J., Baker, J. M., Frankenberg, C., Verma, M., & Yuen, K. (2017). Multiscale analyses of solar-induced florescence and gross primary production. *Geophysical Research Letters*, *44*, 533–541. https://doi.org/10.1002/2016GL070775

Xiao, J., Davis, K. J., Urban, N. M., Keller, K., & Saliendra, N. Z. (2011). Upscaling carbon fluxes from towers to the regional scale: Influence of parameter variability and land cover representation on regional flux estimates. *Journal of Geophysical Research*, *116*, G03027. https://doi.org/10.1029/2010JG001568

Zhang, Y., Joiner, J., Alemohammad, S. H., Zhou, S., & Gentine, P. (2018). A global spatially contiguous solar-induced fluorescence (CSIF) dataset using neural networks. *Biogeosciences*. https://doi.org/10.5194/bg-15-5779-2018

Zoogman, P., Liu, X., Suleiman, R. M., Pennington, W. F., Flittner, D. E., Al-Saadi, J. A., et al. (2017). Tropospheric emissions: Monitoring of pollution (TEMPO). *Journal of Quantitative Spectroscopy and Radiative Transfer*, *186*, 17–39. https://doi.org/10.1016/j.jqsrt.2016.05.008